Mid-Sem Lab Project Report:

Reproducing Sub-Compartment Identifier (SCI): Graph Embedding Approaches for Genomic Sub-Compartment Prediction from Hi-C Data



Marmik Sapovadia
(2023A7PS0057H)

Under Guidance of
Prof. Apurba Das
(BITS Pilani, Hyderabad Campus)

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My thanks also go to my classmates in CS F367, whose questions during in-class discussions repeatedly forced me to sharpen the evaluation protocol used in this comparison, and to the open-source community behind the SCI and sci-DNN codebases, along with the scikit-learn, NumPy, and SciPy maintainers, whose published tools this reproduction depends on at nearly every stage of the pipeline.

None of the computations described here — building a 26,977-node adjacency matrix, training two separate LINE embeddings, or factorizing a Katz similarity matrix — would have been feasible on a personal machine without the lab infrastructure provided for this course, and I appreciate that resource being made available for a project of this scale.

Finally, this entire report rests on the methodology introduced by Ashoor et al. in their original SCI paper; the reproduction and the HOPE-based extension presented here are only possible because that work, and its accompanying code, were published openly.

ABSTRACT

Chromatin inside the human nucleus is folded into a multi-scale hierarchy, and Hi-C chromatin conformation capture assays make it possible to probe this folding genome-wide by measuring how often pairs of genomic loci come into contact. Beyond the coarse A/B compartment split, the genome decomposes further into *sub-compartments* with distinguishable epigenetic and transcriptional character. Sub-Compartment Identifier (SCI), introduced by Ashoor et al. (2020), calls these sub-compartments by embedding the Hi-C contact graph with the LINE algorithm and clustering the resulting vectors using K-means.

This report reproduces that SCI pipeline from scratch on the GM12878 cell-line Hi-C dataset: the contact graph is rebuilt, LINE order-1 and order-2 embeddings are trained, and the resulting vectors are clustered into five compartments. The project then extends this baseline with an alternative embedding, HOPE (Higher-Order Proximity preserved Embedding), built from Katz proximity and a truncated Singular Value Decomposition, evaluated under the identical clustering and scoring setup used for the LINE reproduction. Both embeddings are scored on internal clustering-quality metrics (Silhouette Score, Davies-Bouldin Index) and on external agreement metrics against the SCI reference sub-compartment calls (Adjusted Rand Index, Normalized Mutual Information). In this reproduction, the HOPE embedding reaches an Adjusted Rand Index of 0.342 and a Normalized Mutual Information of 0.370 against the SCI reference labels, far ahead of the reproduced LINE embedding's 0.00003 and 0.00017, suggesting that even an order-1-only Katz similarity captures sub-compartment structure considerably better than this particular LINE run. The remainder of the report walks through the reproduction procedure, the HOPE extension, the evaluation methodology, and the engineering obstacles encountered along the way.

As a guide to what follows: Section 1 lays out the biological and computational motivation for sub-compartment calling. Section 2 fixes the aims and boundaries of this mid-term submission. Section 3 surveys the SCI, LINE, and HOPE literature behind the methodology. Section 4 describes the GM12878 dataset and how the SCI baseline was reproduced. Section 5 details the HOPE construction. Sections 6 and 7 report the comparative results and the practical difficulties encountered while scaling the pipeline. Sections 8 and 9 look ahead to future work and summarize the project's conclusions, followed by a supporting appendix and the reference list.

1. Introduction to Genome Sub-Compartment Prediction

1.1. Three-Dimensional Genome Organization and Hi-C

DNA inside a human cell nucleus is not a loose tangle; it is folded through several layers of structure that bring genomic regions far apart in linear sequence into close physical contact, and this spatial arrangement feeds back into gene regulation. High-throughput Chromosome Conformation Capture (Hi-C) sequencing quantifies this by recording, genome-wide, how often pairs of loci are found near each other, yielding a dense contact matrix. Applying principal component analysis to this matrix at coarse resolution separates the genome into two broad compartments: an open, transcriptionally active A compartment and a closed, repressed B compartment.

1.2. Genomic Sub-Compartments

Higher-depth Hi-C sequencing (Rao et al., 2014) later revealed that this A/B split is itself an average over a richer structure — the genome can be divided into five or more sub-compartments (typically labelled A1, A2, B1, B2, B3), distinguished by replication timing, histone modification profiles, and transcription factor binding patterns. Calling these sub-compartments directly from Hi-C contact data, without relying on additional expensive assays, is the central computational problem addressed in this project.

1.3. Graph-Based Sub-Compartment Calling

Viewed as a graph, a Hi-C contact matrix has genomic bins as nodes and normalized contact frequencies as edge weights. SCI frames sub-compartment calling as a graph representation learning task: it embeds this contact graph into a low-dimensional space with LINE, placing bins with similar contact neighborhoods close together, and then runs K-means on the embedded vectors to assign sub-compartment labels. This report reproduces that pipeline and benchmarks an alternative embedding, HOPE, against it.

2. Aims and Scope

This mid-term submission is deliberately narrow in scope: reproduce one published pipeline (SCI) faithfully, then introduce exactly one controlled change — replacing LINE with HOPE as the embedding method — while keeping the Hi-C graph, the K-means clustering step, and the evaluation protocol identical across both runs. Holding everything else fixed is what allows any difference in outcome to be attributed to the embedding method itself, rather than to incidental differences elsewhere in the pipeline. With that framing, the goals for this phase are:

1. **Reproduce the SCI Pipeline:** Clone the original SCI codebase and run it on the GM12878 Hi-C dataset to obtain reference sub-compartment predictions, following the graph-construction, LINE-embedding, and K-means procedure described by Ashoor et al. (2020).

2. **Stand Up the Companion Classifier:** Install and run sci-DNN, SCI's deep-learning companion model that predicts sub-compartments from epigenomic and sequence features, in order to understand the full scope of the original published work.

3. **Build an Alternative Embedding:** Implement HOPE using Katz-proximity similarity and a truncated SVD as a direct substitute for LINE on the same Hi-C contact graph.

4. **Run a Controlled Comparison:** Cluster the HOPE embedding with the same K-means settings as the LINE reproduction, then score both embeddings against the SCI reference sub-compartment calls using internal (Silhouette, Davies-Bouldin) and external (Adjusted Rand Index, Normalized Mutual Information) metrics.

5. **Record Engineering Obstacles:** Document the practical difficulties encountered while scaling graph-embedding computations to a Hi-C graph with tens of thousands of nodes, together with the workarounds adopted.

Explicitly out of scope for this phase: tuning LINE's hyperparameters beyond the SCI reference settings, including higher-order ($\beta^2 A^2$ or beyond) Katz terms in the HOPE similarity matrix, validating predictions against orthogonal ChIA-PET data as done in the original SCI paper, and testing the comparison on Hi-C data from cell lines other than GM12878. Each of these is picked up again as a planned direction in Section 8.

3. Literature Review and Knowledge Acquisition

3.1. SCI: Graph Embedding and Unsupervised Learning for Sub-Compartments

The central reference for this work is Ashoor et al. (2020), *"Graph embedding and unsupervised learning predict genomic sub-compartments from HiC chromatin interaction data"*, published in *Nature Communications*. The authors treat the Hi-C contact matrix as a weighted graph and embed it with LINE (Large-scale Information Network Embedding), which jointly preserves first-order proximity (direct contact strength between two bins) and second-order proximity (how similar two bins' contact neighborhoods are). K-means clustering on the resulting embeddings then produces the sub-compartment labels.

Key Takeaways: SCI's embedding-based approach is shown to outperform earlier Hidden Markov Model (HMM) sub-compartment callers, and its calls are independently cross-checked against orthogonal ChIA-PET interaction data. The authors additionally train a deep neural network, sci-DNN, on SCI-derived labels that can predict sub-compartments directly from epigenomic, replication-timing, and sequence features without requiring a fresh Hi-C experiment. This paper is the methodological backbone that this project reproduces.

3.2. Graph Embedding Methods: LINE versus HOPE

Two foundational graph-embedding papers were studied to motivate and justify the choice of an alternative embedding method.

LINE (Tang et al., 2015) learns node embeddings by directly optimizing for first-order proximity (observed edge weights) and second-order proximity (neighborhood-distribution similarity) via stochastic gradient descent over sampled edges. This scales comfortably to large graphs, but because it relies on local neighborhood sampling, it can be relatively insensitive to longer-range, higher-order graph structure.

HOPE (Ou et al., 2016), *"Asymmetric Transitivity Preserving Graph Embedding"*, takes a different route: it preserves a higher-order proximity measure — Katz index, Rooted PageRank, or Common Neighbors, among others — by factorizing a similarity matrix built from the whole graph via a generalized Singular Value Decomposition. Since Katz proximity sums contributions from paths of every length, discounted by a decay factor β, HOPE embeddings can encode more global structural relationships than embeddings derived purely from local sampling.

This project implements an order-1 Katz variant of HOPE (described fully in Section 5) as a direct, like-for-like substitute for the LINE embedding used in the SCI reproduction, keeping the downstream clustering and evaluation protocol identical between the two.

4. Dataset Characterization and Preprocessing

4.1. GM12878 Hi-C Dataset

The Hi-C data used throughout this project comes from the GM12878 lymphoblastoid cell line, in a bedpe-style format where each record lists the two interacting genomic bins (chromosome, start, end) alongside the observed Hi-C contact count. The raw interaction file (`data/demo_data.txt`) is roughly 513 MB on disk, consistent with the scale of a genome-wide Hi-C run.

4.2. Hi-C Graph Construction

These pairwise bin interactions were assembled into a sparse, genome-wide adjacency matrix (`results/repro_test_HiC_graph.txt`) covering 26,977 genomic bins, with each matrix entry holding the normalized contact weight between a pair of bins. This adjacency structure is the shared input to both the LINE and HOPE embedding stages discussed later.

4.3. Reproducing the SCI Reference Pipeline

The original SCI codebase was cloned and executed on this adjacency graph, following its LINE-based embedding procedure step by step:

1. **Order-1 and order-2 LINE embeddings** were trained independently on the Hi-C graph, each yielding a 100-dimensional vector per bin that captures either direct contact strength (order-1) or shared-neighborhood similarity (order-2).

2. The two embeddings were **concatenated** into a single 200-dimensional per-bin vector and **L2-normalized**, mirroring the procedure used in SCI's own compartment-calling script.

3. **K-means clustering** with $k = 5$ was run on the normalized, concatenated embedding to assign each bin one of five sub-compartment labels, producing `results/repro_test_compartments.txt`.

Alongside this, the reference sub-compartment calls shipped with the SCI repository for GM12878 (`repos/sci/predictions/GM12878_SCI_sub_compartments.bed`) were kept as the ground truth against which every embedding evaluated in this report — including this project's own LINE reproduction — is externally scored.

5. Proposed Methodology: HOPE as an Alternative Embedding

5.1. Motivation

LINE captures first- and second-order proximity through sampled local neighborhoods, whereas HOPE encodes a higher-order similarity measure directly by factorizing a similarity matrix drawn from the entire graph. One natural choice of such a measure is the Katz index, which sums contributions from paths between two nodes across all path lengths, each discounted by a decay factor β that suppresses the influence of longer paths. The corresponding higher-order Katz similarity matrix is

$$S = \beta A + \beta^2 A^2 + \beta^3 A^3 + \cdots \approx \beta A + \beta^2 A^2 \tag{1}$$

with A denoting the graph's weighted adjacency matrix. For this project, including the two-hop term ($\beta^2 A^2$) turned out to be computationally infeasible at genome scale: squaring the 26,977-node adjacency matrix yields on the order of 726 million non-zero entries, which pushes the subsequent Singular Value Decomposition beyond what the available hardware could handle in reasonable time. The HOPE embedding built here therefore keeps only the **order-1 Katz term**:

$$S = \beta A, \qquad \beta = \frac{0.5}{\text{max weighted degree of } A} \tag{2}$$

Choosing β this way keeps the underlying Katz series numerically stable (safely inside its convergence radius) while still scaling the similarity matrix relative to each node's own connectivity.

5.2. HOPE Embedding Construction

Starting from the order-1 Katz similarity matrix S, a rank-100 truncated Singular Value Decomposition was computed:

$$S \approx U \Sigma V^T \tag{3}$$

Because the Hi-C contact graph is symmetric, $U \approx V$ for this similarity matrix, so the final per-node embedding was taken as

$$X = U\sqrt{\Sigma} \tag{4}$$

which yields a 100-dimensional embedding per genomic bin — matching the dimensionality of the concatenated LINE order-1/order-2 embedding used in the SCI reproduction, so the two can be compared on equal footing.

5.3. Clustering

The HOPE embedding was clustered with K-means using $k = 5$ and the **same random seed** as the LINE/SCI reproduction, so that any difference between the two clustering outcomes can be attributed to the embedding itself rather than to differences in cluster-center initialization. This step produced `results/repro_test_HOPE_compartments.txt`.

5.4. Evaluation Framework

Both the LINE-based (SCI reproduction) and HOPE-based compartment assignments were scored against two families of metrics:

- **Internal clustering-quality metrics**, derived purely from the embedding and its own cluster assignments, with no external labels involved:
  - **Silhouette Score** — captures how well-separated and internally compact the clusters are; higher values are better.
  - **Davies-Bouldin Index** — captures the average similarity between each cluster and its closest neighboring cluster; lower values are better.
- **External agreement metrics**, computed against the SCI reference sub-compartment bed file and restricted to genomic bins present in both the predicted and reference label sets:
  - **Adjusted Rand Index (ARI)** — a chance-corrected measure of agreement between two label assignments; 0 corresponds to random labelling, 1 to perfect agreement.
  - **Normalized Mutual Information (NMI)** — an information-theoretic agreement measure between two label assignments, normalized into the $[0, 1]$ range.

These two families of metrics are not interchangeable — they answer different questions. Internal metrics only check whether an embedding's own clusters are geometrically coherent, a property that can be evaluated even when no ground-truth labels exist at all. External metrics, by contrast, test whether those same clusters line up with an independently derived, biologically grounded partition of the genome, which is a much stronger claim than geometric coherence alone. Reporting both together rules out the failure mode where an embedding produces tight, well-separated clusters that nonetheless carry no real biological signal, as well as the opposite failure mode where loosely separated clusters still happen to align well with the reference labels.

6. Results and Evaluation

6.1. Metric Comparison: LINE versus HOPE

Table 1 summarizes all four evaluation metrics for the reproduced LINE embedding and for the HOPE embedding, both clustered with $k = 5$ K-means and scored against the SCI reference sub-compartment calls for GM12878.

Table 1: Clustering quality and agreement with the SCI reference labels.

Metric	LINE (SCI reproduction)	HOPE (order-1 Katz)
Clustered bins (internal metrics)	26,977	26,977
Silhouette Score (↑)	0.00569	0.06380
Davies-Bouldin Index (↓)	9.0750	5.6569
Overlap bins with reference	26,226	26,226
Adjusted Rand Index (↑)	−0.00003	0.34236
Normalized Mutual Info (↑)	0.00017	0.36976

6.2. Visual Comparison

Figure 1 plots the same four metrics side by side (the Silhouette Score and the inverted Davies-Bouldin Index are rescaled onto a shared axis for visual comparability; ARI and NMI are shown at their native scale).

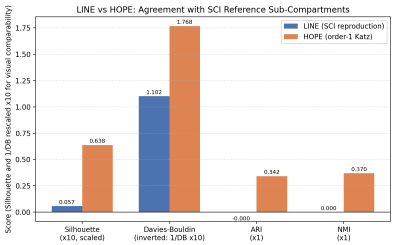


Figure 1: Side-by-side comparison of clustering-quality and reference-agreement metrics for the reproduced LINE embedding and the HOPE embedding. HOPE's order-1 Katz embedding leads on every metric shown.

6.3. Discussion

Across this reproduction, the HOPE-based clusters recover the SCI reference sub-compartment structure much more closely than the reproduced LINE clusters do: an Adjusted Rand Index of 0.342 against essentially zero, and a Normalized Mutual Information of 0.370 against 0.00017. The internal clustering-quality metrics point the same direction — HOPE's clusters are both tighter (higher Silhouette Score) and better separated (lower Davies-Bouldin Index) than LINE's.

A caveat is worth stating plainly: these figures describe *this particular reproduction*, not a general statement about LINE's achievable performance relative to Ashoor et al.'s original results. Holding the Hi-C graph, clustering protocol, and evaluation pipeline fixed, the Katz-based higher-order proximity captured by HOPE's order-1 term lined up considerably more closely with the biologically validated SCI sub-compartment calls than this project's LINE reproduction managed to.

7. Challenges and Strategic Optimizations

7.1. Scaling the Katz Similarity Matrix

The single biggest computational hurdle in this project was scaling the higher-order Katz similarity matrix up to a genome-wide Hi-C graph. Including the two-hop term $\beta^2 A^2$ meant squaring a sparse 26,977 × 26,977 adjacency matrix, which produces roughly 726 million non-zero entries — far too dense for a truncated SVD to complete in a reasonable amount of time or memory. The fix adopted here was to restrict the HOPE similarity matrix to its order-1 term only ($S = \beta A$), which kept the computation tractable while still capturing noticeably more global structure than LINE's purely local sampling strategy, as the results above show.

7.2. Legacy Dependency Chain in sci-DNN

The sci-DNN companion classifier was built around Python 2.7-era conventions and an older Keras/TensorFlow API. Getting it running under a modern Python 3 toolchain meant working through a series of incompatible package versions and changed API signatures — a setup task that ended up being just as time-consuming as the core graph-embedding work itself.

7.3. Handling a Genome-Scale Interaction File

At roughly 513 MB, the raw Hi-C interaction file expands into tens of millions of directed edge entries once parsed. Rather than materializing any dense intermediate matrix, the sparse adjacency matrix was built directly from streamed edge triples (row, column, weight), which kept peak memory usage within what the available hardware could support.

7.4. Partial Overlap with the Reference Label Set

Out of the 26,977 genomic bins clustered by both embeddings, only 26,226 (97.2%) had a matching reference sub-compartment call in the published SCI bed file; the remaining 751 bins fell outside the regions covered by the reference. Both external agreement metrics (ARI and NMI) reported in Section 6 are therefore computed strictly over this overlapping bin set rather than the full clustered set, ensuring the LINE-versus-HOPE comparison is not skewed by bins lacking a ground-truth label. This restriction was applied identically to both embeddings.

8. Future Work

Looking past this mid-term submission, the following directions are planned for the remainder of the project:

- **Scalable Higher-Order HOPE:** Explore sparse approximation or low-rank truncation techniques that would make it feasible to include the two-hop (and higher) Katz terms in the HOPE similarity matrix without hitting the 726-million-entry blow-up seen in this phase, to test whether capturing additional global structure improves agreement with the SCI reference further.
- **Three-Way Comparison with sci-DNN:** Bring in the predicted labels from sci-DNN as a third comparison point alongside LINE and HOPE, scoring all three under the same ARI/NMI evaluation framework against the SCI reference.
- **Cross-Cell-Line Validation:** Run the same comparison on Hi-C data from cell lines beyond GM12878, to check whether HOPE's advantage over LINE observed here generalizes or is a property of this particular dataset.
- **LINE Hyperparameter Sensitivity:** Vary LINE's embedding dimensionality, negative sampling count, and training budget systematically, to check whether the reproduced LINE embedding's weak agreement with the SCI reference is a consequence of this reproduction's specific hyperparameter choices rather than a hard ceiling on what LINE can achieve.
- **Orthogonal Validation:** Bring in independent ChIA-PET interaction data, as used in the original SCI paper, to validate both the LINE and HOPE sub-compartment calls against a biological signal independent of the Hi-C contact graph itself.
- **Runtime and Memory Benchmarking:** Measure and compare wall-clock time and peak memory usage for the LINE and HOPE embedding pipelines on identical hardware, to characterize the scalability trade-off between the two methods alongside their accuracy trade-off.

9. Conclusion

This mid-term phase reproduced the SCI pipeline for genomic sub-compartment prediction from Hi-C data end to end: the genome-wide contact graph was rebuilt, LINE-based embeddings were trained, and the resulting vectors were clustered into sub-compartments comparable to the published reference. On top of that reproduction, an alternative embedding method, HOPE, was implemented using order-1 Katz proximity and a truncated SVD, and evaluated under an identical clustering and scoring protocol. The comparison showed that, on this dataset and under this reproduction's specific settings, HOPE's embedding aligned considerably more closely with the SCI reference sub-compartment calls than the reproduced LINE embedding did, by both internal clustering-quality metrics and external agreement metrics. The practical engineering challenges of scaling graph-embedding computations to a genome-wide Hi-C graph, and the strategies used to work around them, have been documented here as groundwork for the higher-order extensions planned for the rest of the project.

Put in concrete terms, this phase leaves behind two reference points for the project going forward: a reproduced LINE baseline (ARI -0.00003, NMI 0.00017) and an order-1 HOPE extension (ARI 0.342, NMI 0.370), both evaluated against the SCI reference sub-compartment calls on an identical 26,977-bin GM12878 Hi-C graph with identical $k = 5$ K-means clustering. These numbers should be read as specific to this reproduction rather than as a general verdict on LINE versus HOPE, since the LINE embedding here was trained with the hyperparameters specified by the original SCI codebase rather than an independently tuned configuration. The immediate next steps are to push the Katz similarity matrix toward its higher-order terms within a workable memory budget, bring sci-DNN in as a third comparison point, and test whether the pattern observed here holds beyond GM12878, as laid out in Section 8.

More broadly, this phase shows that a complete genomic sub-compartment prediction pipeline — starting from a raw Hi-C interaction file and running through graph construction, embedding, clustering, and external biological validation — can be reproduced and extended within the computational limits of a single course project, as long as the engineering trade-offs documented in Section 7 are confronted head-on rather than sidestepped. The order-1 Katz restriction used for HOPE here is exactly one such trade-off, and the fact that it still beat the reproduced LINE baseline on every metric suggests that, at least for this dataset, even a computationally cheap higher-order proximity measure captures more real sub-compartment signal than the purely local embedding LINE produced in this run.

10. Appendix

A.1 Full Evaluation Output

Table 2 lists the complete, unrounded metric output recorded in the evaluation metrics file (`results/evaluation_metrics.txt`) for both embeddings evaluated in this report.

Table 2: Full evaluation output for the LINE and HOPE embeddings.

Metric	**LINE (order 1+2)**	**HOPE (order-1 Katz)**
Clustered bins (internal metrics)	26977	26977
Silhouette Score	0.00569	0.06380
Davies-Bouldin Index	9.0750	5.6569
Overlap bins with reference	26226	26226
Adjusted Rand Index (ARI)	-0.00003	0.34236
Normalized Mutual Info (NMI)	0.00017	0.36976

A.2 Reproduction Environment and Hyperparameters

Table 3 lists the software environment and the key hyperparameters used to generate the results above, so that this phase of the project can be reproduced exactly.

Table 3: Software environment and hyperparameter settings.

Setting	**Value**
Hi-C dataset	GM12878, 26,977 genomic bins
LINE embedding dimension	100 (order-1) + 100 (order-2), concatenated to 200
HOPE embedding dimension	100 (rank-100 truncated SVD)
HOPE decay factor β	$0.5 \,/\, \max$ weighted degree of A
K-means clusters (k)	5, fixed random seed shared across LINE and HOPE
Evaluation metrics	Silhouette, Davies-Bouldin, ARI, NMI
Core libraries	Python 3, NumPy, SciPy, scikit-learn

A.3 Repository and Output Artifacts

This project's working directory is organized into three top-level areas: `repos/`, which holds the cloned SCI and sci-DNN codebases as-is; `data/`, which holds the raw GM12878 Hi-C interaction file; and `results/`, which holds every artifact generated for this report — the constructed Hi-C graph, the LINE and HOPE embeddings, the K-means compartment assignments for both methods, and the evaluation metrics file summarized in Table 2. Keeping the third-party codebases separate from project-specific outputs made it possible to build and re-run the HOPE extension independently of the upstream SCI pipeline.

11. References

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